

RICHARD LOURETTE

Principal Embedded Software Engineer

 flourette@gmail.com |  1.585.953.5309 |  Rochester/Oswego, New York

 Available for Remote Work

PROFESSIONAL SUMMARY

Principal Embedded Software Engineer with 30+ years of experience architecting, developing, and optimizing embedded software solutions for mission-critical systems. Expert in C/C++ development, embedded Linux, RTOS, and bare-metal programming with proven leadership in technical architecture and team mentorship. Most recently led PTP/gPTP (IEEE 802.1AS) Grandmaster development at Point One Navigation, spanning STM32L4 MCU firmware, iMX8 embedded Linux host software, and PHC synchronization infrastructure, while driving a major re-architecture of the platform's services layer to improve modularity, testability, and firmware/host independence. Demonstrated success delivering scalable embedded systems for aerospace, defense, automotive, and IoT applications with emphasis on performance optimization and safety-critical requirements.

Core Competencies: Embedded Systems Architecture · C/C++ · Python · C# · Real-Time Systems · Embedded Linux · RTOS · PTP/gPTP & Precision Timing · GNSS Integration · Hardware-Software Integration & Co-Design · Services Architecture · Performance Optimization · Fault-Tolerant Design · Technical Leadership · Consulting

TECHNICAL EXPERTISE

Programming Languages: C/C++ (30+ years), Python, C#, Java, Assembly Language, Shell Scripting, HTML/CSS, Forth

Embedded Platforms: ARM Cortex-A/M/R, STM32L4 (Cortex-M4), iMX8, TI ARM A9, DSP processors, Zynq UltraScale+, Versal ACAP, 8/16/32-bit microcontrollers, Radiation-hardened MCUs

Operating Systems: Embedded Linux, BalenaOS, FreeRTOS, RTEMS RTOS, Nucleus/Nucleus++ RTOS, Bare Metal Programming

Communications & Timing Protocols: IEEE 802.1AS (gPTP), IEEE 1588 (PTP), SpaceWire, SpaceFibre, CAN/CAN-FD, I²C, SPI, UART, Ethernet, TCP/UDP, RESTful APIs

Development Tools: GCC Toolchain, GDB, JTAG Debuggers, Git, CI/CD Pipelines, Jenkins, Docker (custom base images, cross-compilation in containers, CI pipeline integration, BalenaOS service deployment), Bazel

Performance & Validation Tools: perf, Valgrind, Flame Graphs, Wireshark/tshark, AF_NETLINK, linuxptp/ptp4l, Static Analysis Tools

Testing Frameworks: Google Test, Unity, Pytest, Hardware-in-the-Loop Testing, CI/CD-integrated test pipelines, Automated White-Box Testing

Hardware Integration: FPGA/SoC co-design, PCB collaboration, PHC synchronization, 1PPS timing, containerized embedded service deployment via Docker and BalenaOS, signal processing, sensor fusion, power optimization

Methodologies: Agile/Scrum, Test-Driven Development (TDD), code reviews, requirements analysis, system architecture, high-assurance practices

Key Domains: Spacecraft flight software · GNSS receivers · PTP/gPTP timing systems · Real-time 3D imaging · Biometric wearables · Industrial IoT · Airborne surveillance · High-reliability & fail-operational systems

PROFESSIONAL EXPERIENCE

POINT ONE NAVIGATION | Staff Embedded Firmware Engineer

January 2026 – Present | Remote

- Led PTP/gPTP (IEEE 802.1AS) Grandmaster support on the Atlas GPS/PTP timing platform, working across both STM32L4 MCU firmware (FreeRTOS C++) and embedded Linux host software (C++/Python, BalenaOS)
- Implemented GNSS-disciplined PTP Hardware Clock (PHC) synchronization on the iMX8 Linux host, including 1PPS GPIO timestamping on the STM32 MCU, AF_NETLINK-based PHC watchdog recovery from link-bounce resets, and IEEE 802.1AS gPTP profile configuration via REST API
- Driving a major re-architecture of the Atlas Services layer, transitioning Septentrio GNSS receiver initialization from MCU to host and introducing asynchronous command/response processing with a shared I/O worker model, improving modularity, testability, and the ability to evolve host and firmware independently
- Refactored Septentrio GNSS receiver initialization with a robust six-phase sequence covering baud-rate auto-detection, RTCM escape handling, SBF stream configuration, and non-volatile config save
- Built a PTP test bench using the Dart platform to calibrate and validate the Grandmaster design; developed custom validation tools alongside tshark/Wireshark to verify protocol correctness and timing accuracy on the wire
- Contributed upstream BalenaOS pull requests for device tree customization to support Atlas hardware requirements
- Established Linux cross-build infrastructure for the STM32 firmware and maintained Google Test-based CI test suites, enabling reliable automated validation across firmware and host software changes

RL TECH SOLUTIONS LLC | President & Chief Technology Officer

October 2022 – Present | Rochester, NY

D3/L3Harris Aerospace – Chief Engineer Consultant

Oct 2022 – Dec 2023; Oct 2024 – Present

- Designed spacecraft payload systems integrating 5 radiation-hardened MCUs via SpaceWire communication
- Architected embedded software using NASA Core Flight System (cFS) on RTEMS RTOS
- Implemented microservices architecture to enhance reliability, maintainability, and modularity of flight software
- Led technical design reviews and mentored engineers on embedded systems and space-qualified software practices
- Authored and presented gate review, contributing to award of eight-figure payload contract

- Returned post-sabbatical to develop embedded software architecture for advanced real-time time-of-flight 3D imaging system

Topcon Positioning Systems – Senior Embedded Software Consultant

October 2023 – April 2025

- Architected and implemented Linux C++ subsystems for next-generation GNSS receivers on embedded ARM A9 TI processors
- Built high-performance multi-threaded applications optimized for battery-powered, resource-constrained devices
- Delivered 150,000+ lines of production C++ code with full test coverage using Python, pytest, and Google Test
- Designed advanced white box testing strategies with systematic source analysis for comprehensive test plan generation
- Improved system performance by 40% through CPU profiling with perf and flame graph analysis
- Developed custom loopback filesystem and Python curses monitoring tools for real-time CPU/thread utilization debugging
- Utilized modern Git workflows and CI/CD pipelines aligned with industry best practices to ensure high-quality, reliable software delivery

Professional Sabbatical *May 2025 – Sep 2025*

- Focused on renewal and independent research, returning with published articles and refreshed technical perspective

PANASONIC INDUSTRIAL IOT DIVISION | Engineering Group Manager

February 2021 – October 2022 | Rochester, NY

- Managed cross-functional engineering teams spanning RF Engineering, protocol design, antenna development, mesh networking, and embedded firmware
- Directed development of industrial IoT devices with integrated RESTful interfaces for enterprise system integration
- Resolved critical RF communication protocol issues for 2000+ device customer deployment during COVID-19 constraints
- Transformed testing processes from manual to fully automated, reducing test cycle time from weeks to 3 days
- Led technology roadmap development for next-generation industrial IoT product portfolio

TOKENIZE INC. | Vice President of Engineering

September 2015 – February 2021 | Rochester, NY

- Spearheaded R&D for biometric wearable devices incorporating NFC technology and capacitive fingerprint scanning
- Designed ultra-low-power embedded electronics meeting strict battery life requirements for wearable applications
- Developed real-time signal processing algorithms and visualization tools for capacitive sensing systems

- Collaborated with hardware teams on power optimization strategies for resource-constrained wearable devices

L3HARRIS GEOSPATIAL SYSTEMS | Chief Scientist/Principal Investigator

May 2002 – September 2015 | Rochester, NY

- Developed advanced in-situ telescope wavefront correction algorithms using Python for space-based optical systems
- Served as lead architect for GPS-III satellite program test equipment, defining system requirements and FPGA-based hardware architecture
- Designed onboard payload processing electronics for Wide Area Airborne Surveillance (WAAS) systems
- Led development of 7-slot high-altitude VPX supercomputing cluster for Advanced Geospatial Processing applications
- Architected electronic payload systems for visual and infrared sensor platforms

EASTMAN KODAK COMPANY | Chief Firmware Architect

1995 – 2002 | Rochester, NY

- Led international firmware development teams for hybrid consumer digital camera products
- Integrated emerging sensor technologies and wireless capabilities (Bluetooth) into embedded camera systems
- Designed application frameworks and device drivers for memory-constrained embedded systems
- Co-architected USB device drivers and PIMA 15740 application layer protocol implementation
- Developed real-time image processing algorithms for laser printer marking systems
- Created calibration algorithms for non-linear laser marking engine components

KEY ACHIEVEMENTS

- **Patent Portfolio:** 20+ issued US patents plus international patents in embedded systems and image/signal processing
- **Performance Optimization:** Achieved 40% gains via systematic profiling and refactoring
- **Team Leadership:** Successfully managed and mentored 13+ engineers across multiple disciplines
- **Process Improvement:** Reduced verification cycles from weeks to days through automation implementation
- **High-Reliability Systems:** Delivered fault-tolerant, mission-essential embedded software for aerospace and defense programs
- **Technology Innovation:** Led adoption of emerging technologies in embedded systems across multiple industries
- **Contract Win:** Secured contract totaling \$50M+ through technical leadership, customer engagement, and architecture excellence

EDUCATION

Bachelor of Science in Electrical Engineering

University of Dayton | Dayton, Ohio

SECURITY CLEARANCES

- Previously held DoD Top Secret Clearance
- Completed Single Scope Background Investigation (SSBI) for Sensitive Compartmented Information (SCI) access
- Available for security clearance reinstatement

ADDITIONAL QUALIFICATIONS

- **Amateur Radio License:** FCC Amateur Extra Class license holder (AB2MD)
- **Remote Work Experience:** 5+ years of successful remote collaboration and team leadership
- **Consulting Expertise:** Proven track record as technical consultant for Fortune 500 companies
- **Industry Recognition:** Subject matter expert in embedded systems with 30+ years of deep technical expertise
- **Continuous Learning:** Active in embedded systems community and emerging technology adoption